

Securing Data and Advancing Sustainability on AWS with IU International University of Applied Sciences

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—Kevin Fischer, Head of DevOps



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Imagine trying to study for a big exam without reliable access to digital program materials. Disheartening doesn't even begin to cover it. That's why Germany's largest university by student enrollment, [IU International University of Applied Sciences](#) (IU), is set on providing highly reliable service to its more than 100,000 students. The university wants to provide a modern education experience that makes it easy for students to



get clarity on assignments, anytime and anywhere. And while it grows to meet global demand for high-quality education, the university wants to verify that it scales securely and sustainably.

IU kicked off a migration to Amazon Web Services (AWS) in 2020 to further its mission of improving access to education for everybody, everywhere, at any time. Data security and compliance were also major reasons for the migration. “Speed, flexibility, scalability, security, compliance, and reliability are the biggest benefits of our migration to AWS,” says Kevin Fischer, head of DevOps at IU. Using AWS, the university has been certified as carbon neutral by TÜV SÜD for 3 consecutive years, making IU one of the world’s first climate-neutral distance learning universities.

Minimizing Carbon Emissions on AWS for IU

IU is a private, state-accredited university with more than 35 locations in Germany and a global student body. Although it offers in-person courses, most of IU’s students are distance learners. In fact, IU estimates that it [reduces greenhouse gas emissions by seven times](#) when it teaches a distance learner as opposed to a regular offline student. But the university wants to do more than just teach about sustainability—it wants to incorporate environmentally responsible practices across all areas of operation. “Adopting the AWS Cloud is not just a technology upgrade; it is directly linked to our sustainability strategy. By harnessing the power of the cloud, we can reduce our environmental footprint, optimize our use of resources, and move toward a more sustainable future where efficiency meets ecology,” says Fischer. “We host our systems in the Europe (Frankfurt) AWS Region, which is powered by over 95 percent renewable energy, to unlock low latency while minimizing emissions.”

Prior to using AWS, the majority of IU’s software was hosted by an external provider. However, as IU grew, it needed to transform its technology to scale sustainably and securely. In July 2020, the university built a dedicated DevOps team and rearchitected its systems on AWS. IU mapped out a clearly structured plan and defined the security protocols that it wanted to put in place to protect student data. With the average cost of a ransomware attack hitting \$1.5 million, security is a high priority for IU. “Research universities are often targeted by cyberattacks,” says Fischer. “We work alongside AWS to implement simple-to-manage security solutions.”

Improving Student and University Data Using Amazon ECS

IU is committed to innovating the education experience for their students. Its teams enjoy experimenting with new services and solutions to improve quality and provide new functionality. “On our old system, teams used to wait 8 to 9 days for new servers to spin up when they wanted to try a new service,” says Fischer. “Now, we just click a button, and we can start working within a few minutes.” The migration to AWS has advanced the agility of IU teams and increased the pace of innovation.

By using infrastructure as code on AWS, IU has done more than increase agility—it has fortified the security of student and university data. Using AWS security tools, IU can verify that misconfigurations do not impact its cloud systems. “On AWS, we wanted to experiment freely with new ideas and features without being encumbered by hardware constraints,” says Sebastian Glöckner, chief technology officer at IU. “More importantly, we wanted to verify that everything we built is secure.”

Using Purpose-Built AWS Services to Create Flexible Systems

The university uses 38 different AWS services. IU runs its infrastructure on containers, and it uses Amazon Elastic Container Service (Amazon ECS), a fully managed container orchestration service, to simplify its architecture. It also advances simplicity by using Amazon Elastic Kubernetes Service (Amazon EKS), a managed service that runs Kubernetes. “Amazon ECS and Amazon EKS are the heart of our infrastructure because every application runs on them,” says Fischer.



As the fastest-growing university in Europe, IU promotes a student-centered mindset across its operations. Using AWS advances the organization's goal of improving accessibility to education for people all over the world. "On AWS, we don't have to worry about where our students live and study," says Glöckner. "By using AWS services, we offer our students a highly personalized learning experience by delivering a high-quality, user-centered, and scalable digital learning environment."

Although its IT workforce has grown to about 300 people, IU relies on only eight DevOps employees to handle its infrastructure. The university uses managed services and automation wherever it can to reduce workloads and accelerate the pace of innovation. "Using AWS, we pay only for what we use," says Fischer.

IU highlights a number of key takeaways from its digital transformation journey. The university now embraces automation to streamline processes and reduce the potential for human error. It also welcomes opportunities to imbue its systems with flexibility. "On AWS, we have a much more resilient, high-performing system that operates economically and empowers our staff to work more efficiently," says Fischer.

Using AI to Enhance the Student Experience on AWS

With a more secure, carbon-neutral infrastructure in place, IU is ready to accelerate the pace of innovation. The university is building artificial intelligence (AI) products to answer student questions about course requirements and content. "We are building more teams to deliver more products," says Fischer. "We have four teams working AI right now, and we're building three more as we speak."

IU plans to continue adopting new services and solutions as it builds out its next generation of student offerings. "Our AWS account team is always informing us about changes and helping us solve challenges," says Fischer. Together, the organizations are democratizing education through technology.

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